Introduction to Convergent Technologies

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Convergent Technologies (A29442)

Short Title: Int to Convergent Technologies
Faculty Science and Computing
Department Engineering Technology
Cluster Engineering
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Credits 5 **Level:** Advanced

Description of Module / Aims

This module aims to build on the students undergraduate knowledge and introduce him / her to the state-of-the-art in convergent technologies for advanced bio-medical and electro-mechanical applications. A selection of convergent technologies for advanced bio-medical and electro-mechanical applications and industry roadmaps are presented and examined in the context of their associated regulatory frameworks.

Programmes

TECH-0048	BSc (Hons) in Applied Computing (WD_KACCM_B)	4 / 7 / E
TECH-0048	BSc (Hons) in Applied Computing (WD_KCOMP_B)	4 / 7 / E
TECH-0048	BSc (Hons) in Computer Science (WD_KCMSC_B)	4 / 7 / E
TECH-0048	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 7 / M
TECH-0048	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	4 / 7 / M

Indicative Content

- Overview of the concept of convergent technologies.
- Review of typical industry driven technology convergence roadmaps.
- Review of key regulatory frameworks and application driven ethical considerations.
- Review of typical bio-mechanical measurement variables.
- System and block level review of available technologies for data handling, signal monitoring and waveform generation in advanced bio-medical and electro-mechanical applications
- Advanced Case Studies (Ex.: structural health monitoring, pulse oximeter, ECG, pacemaker etc.).

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate a knowledge and comprehension of the fundamental importance of convergent technologies in the development of advanced bio-medical and electro-mechanical applications.
2. Demonstrate a knowledge and comprehension of basic bio-mechanical measurement variables, signal monitoring / waveform generation and data handling technologies that are commonly employed in advanced bio-medical and electro-mechanical applications.
3. Demonstrate an ability to apply the knowledge and comprehension gained in analysing advanced bio-medical and electro-mechanical applications and able to relate system specifications to implementation technologies.
4. Demonstrate ability to develop system and block level design criteria and partial solutions for selected bio-mechanical applications.
5. Demonstrate an understanding of associated regulatory and ethical considerations.

Learning and Teaching Methods

- Lectures

- Case Studies
- Mini Projects
- Presentations

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Case Studies	50%	1,2,3,4,5
Assignment	50%	1,2,3,4,5

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	12	
Independent Learning	99	

Essential Material

- "Institute of Electrical and Electronic Engineers." www.ieee.org

Supplementary Material

- Bronzion, J.D. _Medical Devices and Systems_. .: CRC Press, 2006.
- Brown, B.H., R.H. Smallwood, D.C. Barber, P.V. Lawford and D.R. Hose. _Medical Physics and Biomedical Engineering_. .: Institute of Physics, 1999.
- Chan, A.Y.K. _Biomedical Device Technology: Principles & Design_. .: Thomas, 2008.
- Enderle, J., S.M. Blanchard and J. Bronzino. _Introduction to Biomedical Engineering_. .: Academic Press, 2005.
- Hobbie, R.K. _Intermediate Physics for Medicine & Biology_. .: Springer - Verlag, 1997.
- Khandpu, R. _Bio-Medical Instrumentation: Technology & Applications_. UK: McGraw Hill, 2004.
- Prutchi, D. _Design & Development of Medical Electronic Instrumentation_. UK: Wiley Interscience, 2004.
- Stree, L.J. _Introduction to Biomedical Engineering Technology_. UK: CRC Press, 2007.
- Togawa, T. _Bio-Medical Transducers and Instruments_. UK: CRC Press, 1997.
- Webster, J.G. _Medical Instrumentation Application and Design_. UK: Houghton Wiley, 1999.